

Lecture 31

STAT 109: Introductory Biostatistics

Lecture 31

Matched pairs

In a matched pairs design, each unit contributes two measurements that belong together (same person at two times, two body sides, twins, and so on). Pairs are analyzed using within-pair differences (or tests that treat pairing correctly) because the two values in a pair are not independent.

Example 1: Grip strength, dominant vs non-dominant hand

Hypothetical grip strength (kg) for the same eight students, measured once per hand.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {  
  install.packages("tidyverse", repos = "https://cloud.r-project.org")  
}
```

```
library(tidyverse)
```

```
grip <- tibble(  
  student = paste0("S", 1:8),  
  dominant_kg = c(42, 38, 45, 41, 36, 44, 39, 43),  
  nondominant_kg = c(38, 35, 42, 37, 33, 40, 36, 40)  
) |>  
  mutate(diff_d_minus_nd = dominant_kg - nondominant_kg)
```

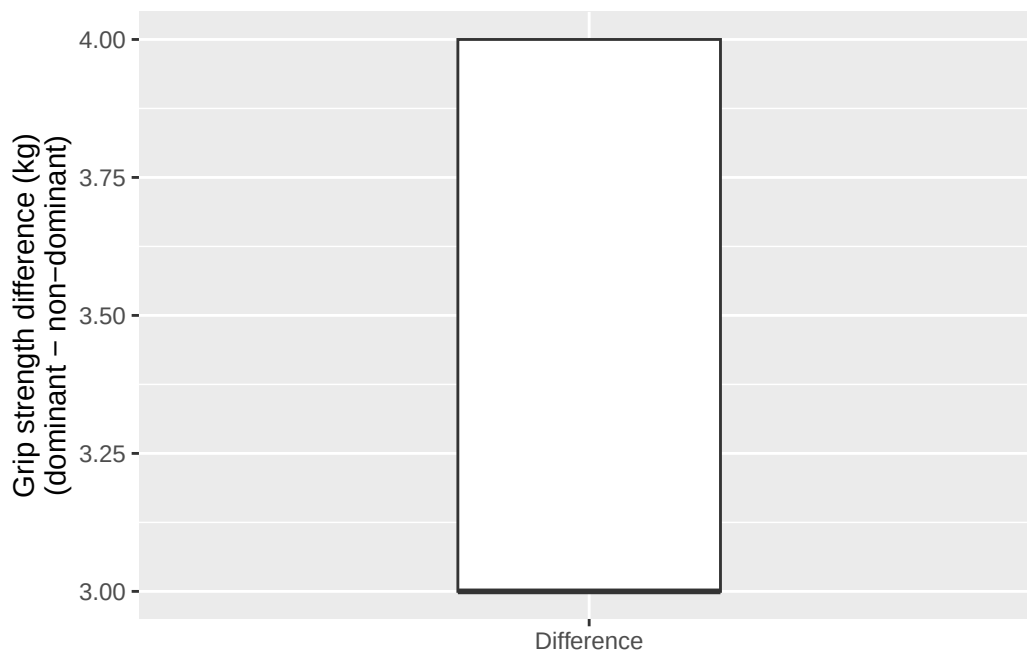
```
grip
```

```
# A tibble: 8 x 4  
  student dominant_kg nondominant_kg diff_d_minus_nd  
  <chr>         <dbl>         <dbl>         <dbl>  
1 S1             42             38             4  
2 S2             38             35             3  
3 S3             45             42             3  
4 S4             41             37             4  
5 S5             36             33             3  
6 S6             44             40             4  
7 S7             39             36             3  
8 S8             43             40             3
```

Boxplot of within-person differences (dominant — non-dominant)

```
ggplot(grip, aes(x = "Difference", y = diff_d_minus_nd)) +  
  geom_boxplot(width = 0.35) +  
  labs(  
    x = NULL,
```

```
y = "Grip strength difference (kg)\n(dominant - non-dominant)"
)
```



Paired *t*-test: 95% confidence interval and test of zero mean difference

H_0 : the **population mean** of (dominant – non-dominant) grip strength is 0.

```
t.test(
  grip$dominant_kg,
  grip$nondominant_kg,
  paired = TRUE,
  conf.level = 0.95
)
```

Paired *t*-test

```
data:  grip$dominant_kg and grip$nondominant_kg
t = 18.445, df = 7, p-value = 3.413e-07
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 2.942318 3.807682
sample estimates:
mean difference
      3.375
```

The same result comes from a one-sample *t*-test on the differences:

```
t.test(grip$diff_d_minus_nd, mu = 0, conf.level = 0.95)
```

One Sample *t*-test

```
data:  grip$diff_d_minus_nd
```

```
t = 18.445, df = 7, p-value = 3.413e-07
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 2.942318 3.807682
sample estimates:
mean of x
 3.375
```

The 95% confidence interval is for that population mean difference; the p -value summarizes evidence against H_0 .

Example 2: Paired proportions — McNemar's test

For **two binary outcomes on the same people** (e.g. before vs after), use **McNemar's test** (`mcnemar.test()`). A two-sample `prop.test()` is for **independent** groups and is not appropriate here.

Ten hypothetical students: whether they meet a daily step goal in week 1 (1=yes, 0=no) and again in week 2.

```
steps <- tibble(
  student = paste0("S", 1:10),
  week1 = c(0, 0, 0, 0, 1, 1, 0, 0, 1, 1),
  week2 = c(0, 0, 1, 1, 1, 0, 1, 0, 1, 1)
)
```

```
steps
```

```
# A tibble: 10 x 3
  student week1 week2
  <chr>    <dbl> <dbl>
1 S1      0      0
2 S2      0      0
3 S3      0      1
4 S4      0      1
5 S5      1      1
6 S6      1      0
7 S7      0      1
8 S8      0      0
9 S9      1      1
10 S10     1      1
```

Cross-classify the paired outcomes (rows = week 1, columns = week 2; 0 = no, 1 = yes):

```
tab_steps <- table(week1 = steps$week1, week2 = steps$week2)
tab_steps
```

```
      week2
week1 0 1
0 3 3
1 1 3
```

```
mcnemar.test(tab_steps)
```

McNemar's Chi-squared test with continuity correction

```
data:  tab_steps
McNemar's chi-squared = 0.25, df = 1, p-value = 0.6171
```

McNemar tests whether the population proportions of “yes” differ between the two occasions for paired binary data using only the **discordant** pairs ($0 \rightarrow 1$ vs $1 \rightarrow 0$). A small p-value is evident against the null hypothesis of no different in proportions between the two measurement times.